

Lecture 2 — Finite Differences and Boundary Conditions

18.085 Computational Science and Engineering I • MW 9:30–11, 6-120 • Monday, September 14, 2026

After this lecture you should be able to

- Derive the equilibrium equation $-Cu'' = f$ for an elastic rod from a force balance on one small segment.
- Discretise $-u'' = f$ on a finite grid and read off the interior rows of the resulting matrix.
- Turn fixed, free, and periodic end conditions into boundary rows, producing $\mathbf{K}, \mathbf{T}, \mathbf{B}, \mathbf{C}$.
- Exhibit the nullspace of $\mathbf{B}$ and say what it means physically.

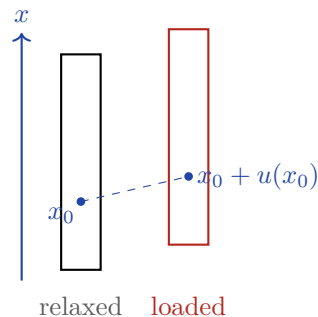
Where we left off

Last time we built the second-difference operator as an *infinite* matrix, and then admitted that this is a fiction: a real computation happens on a finite grid over a finite interval, and the grid has two ends. Today we truncate. What we will find is that the interior of the matrix is the easy part — the $1, -2, 1$ stencil, unchanged — and that all of the modelling lives in the first and last rows.

To decide what those rows should be, we need a physical problem, because a mathematical interval does not tell you what is happening at its endpoints. A rod does.

1 A physical example: an elastic rod under tension

Take the one-dimensional deformation of a light elastic rod. The object we are solving for is the **displacement field** $u(x)$: how far the slice of the rod that sits at coordinate x when the rod is relaxed moves once external forces are applied. The slice that started at x_0 ends up at $x_0 + u(x_0)$.



Goal. Given the distribution of forces along the rod, $f(x)$, calculate the resulting displacement $u(x)$.

A linear stress–strain relation

Assume the material responds linearly. Look at a tiny segment between x and $x + \Delta x$. It has been stretched by $u(x + \Delta x) - u(x)$, and the tension it carries is proportional to that stretch *per unit length*:

$$T(x) = C \frac{u(x + \Delta x) - u(x)}{\Delta x}.$$

Taking $\Delta x \rightarrow 0$, the displacement $u(x)$ induces a tension in the rod given by

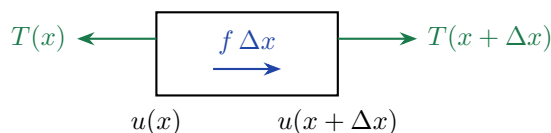
$$T(x) = C u'(x).$$

What C hides

C is a “tension constant” — Young’s modulus for the material combined with the cross-sectional area of the rod, and whatever else is needed to convert a strain into a force. For now it is a single number, and we will set it to 1. It matters that it sits *between* the two derivative steps below; when we return to this structure in the springs-and-masses lecture, that middle slot is where the material properties live.

2 Force balance gives the equilibrium equation

Now introduce the external forces and draw the free-body diagram of a small segment. Three things act on it: the tension pulling out at the right face, the tension pulling out at the left face, and the applied load f spread over a length Δx .



Nothing is accelerating — this is a statics problem — so the three contributions sum to zero:

$$T(x + \Delta x) + f \Delta x - T(x) = 0 \quad \implies \quad - \left(\frac{T(x + \Delta x) - T(x)}{\Delta x} \right) = f(x).$$

In the continuous limit this is $-T'(x) = f(x)$, and substituting $T = Cu'$ gives the equation we will be discretising for the rest of the lecture:

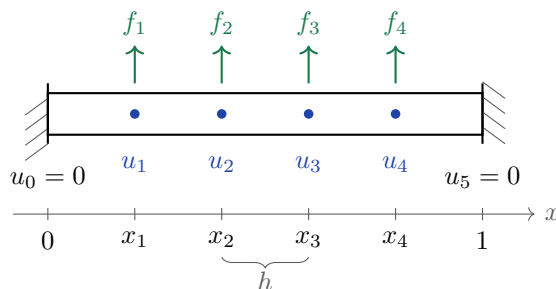
$$-C u''(x) = f(x)$$

Two first-order steps, not one second-order step

The chain is: displacement $\xrightarrow{\text{differentiate, } \times C}$ tension $\xrightarrow{\text{differentiate, negate}}$ load. The second derivative is not primitive here; it is the composition of two first differences with the material constant in between. This is the factored structure promised at the end of Lecture 1, and keeping it visible is what will eventually make the matrices symmetric.

3 Fixed–fixed ends: the matrix K

Let us discretise using finite differences, starting with **fixed** boundary conditions. Take $C = 1$. The elastic rod is fixed between $x = 0$ and $x = 1$, and neither end of the rod can be displaced.



The grid has spacing h , the nodes are $x_j = jh$, and the two end values u_0 and u_5 are *known* — both are zero. The unknowns are the four interior values u_1, u_2, u_3, u_4 , and the data are the loads $f_1, \dots, f_4$ at those nodes.

Using the three-point centred difference scheme for u'' from last lecture,

$$-u''(x) \simeq \frac{-u(x-h) + 2u(x) - u(x+h)}{h^2},$$

and given our discrete grid, this translates to

$$\frac{1}{h^2} \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}.$$

Where each row came from

The **interior rows** — rows 2 and 3 — follow directly from the finite difference formula. Nothing happens to them; they are the infinite matrix of Lecture 1, copied.

The **boundary rows** are the content. Take the first one. We have $u_1 = u(h)$, $u_2 = u(2h)$, and $u_0 = u(0) = 0$ because the end is fixed. So

$$-u''(h) \simeq \frac{-u_0 + 2u_1 - u_2}{h^2} = \frac{1}{h^2}(2u_1 - u_2),$$

and the same argument at the other end, using $u_5 = 0$, produces the last row.

The corner entry is not a truncation

The u_0 term did not vanish because we ran out of grid and threw it away. It vanished because u_0 is *known* and its known value happens to be zero. Keep the two facts separate in your head — which nodes are unknowns, and what the known ones equal — and the boundary rows stop being mysterious.

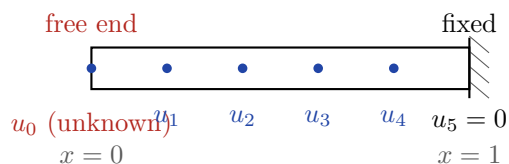
This is the first of the four special matrices we will use all semester:

$$\mathbf{K} = \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix}, \quad \text{fixed-fixed, and the system is } \frac{1}{h^2} \mathbf{K} \mathbf{u} = \mathbf{f}.$$

Throughout the course the letters $\mathbf{K}, \mathbf{T}, \mathbf{B}, \mathbf{C}$ name the *integer* arrays; the factor $1/h^2$ rides outside.

4 A free end: the matrix T

Now release the end at $x = 0$ and keep the end at $x = 1$ fixed.



At the free end there are no constraints on the rod, so there is no internal tension coming in there:

$$T(x=0) = 0 \quad \implies \quad \boxed{u'(0) = 0}.$$

Two things change. First, since $x = 0$ is now a free end, u_0 is an **unknown to be solved for** rather than a given value — the system grows by one row and one column, and the load f_0 at that node joins the right-hand side. Second, $u_5 = 0$ is still known; that is the fixed condition at the far end. The finite difference discretisation now reads

$$\frac{1}{h^2} \begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} f_0 \\ f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \quad \mathbf{T} = \begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}.$$

The first row is the one that corresponds to the free end. Here we take the second difference one-sidedly, as a difference of first derivatives, and then use the boundary condition:

$$-u''(0) \simeq \frac{u'(0) - u'(h)}{h} = -\frac{u'(h)}{h} = -\frac{1}{h} \left(\frac{u_1 - u_0}{h} \right) = \frac{1}{h^2} (u_0 - u_1),$$

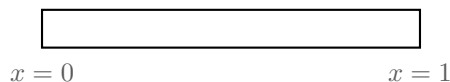
where the middle equality is exactly the statement $u'(0) = 0$ imposed by the free end. That is the row $[1 \ -1 \ 0 \ 0 \ 0]$.

One entry records the physics

Fixing an end puts a 2 in the corner; freeing it puts a 1. The entire difference between “held” and “released” at that end is one number in one corner of the matrix. Everything else — the interior stencil, the grid, the load vector — is untouched.

5 Free–free: the matrix B , and why it is singular

Release both ends. The rod now floats.



The boundary conditions now read

$$u'(0) = 0, \quad u'(1) = 0,$$

and the same one-sided argument as above, applied at both ends, gives a 1 in each corner. The resulting finite difference matrix is

$$\frac{1}{h^2} \begin{bmatrix} 1 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 1 \end{bmatrix}.$$

This matrix is **non-invertible**, or singular. It has a non-trivial nullspace: every vector

$$\mathbf{u} = a \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{solves} \quad \mathbf{B}\mathbf{u} = \mathbf{0},$$

as you can check row by row — each row of $\mathbf{B}$ sums to zero, so a constant vector is annihilated by all four of them.

Physically, these solutions correspond to **shifting the rod uniformly in space**, which is allowed precisely because the rod is free at both ends. Nothing is holding it anywhere, so if u is an equilibrium displacement then so is u plus a constant: slide the whole rod one centimetre to the right and every internal tension is unchanged.

Invertibility is a statement about physics

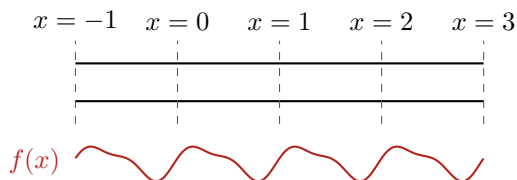
$\mathbf{K}$ is invertible and $\mathbf{B}$ is not, and the reason is not arithmetic. A rod held at both ends has exactly one equilibrium for a given load, so the map from displacement to load can be run backwards. A rod held at neither end has no preferred position at all, so that map cannot be run backwards — and the ambiguity is not vague, it is the explicit vector $(1, 1, 1, 1)$. The nullspace of a matrix is a list of the things your model failed to pin down.

The other half of the question

If $\mathbf{B}$ cannot be inverted, which loads $\mathbf{f}$ can a free-free rod support at all? That is the question of whether $\mathbf{f}$ lies in the space of vectors reachable as $\mathbf{B}\mathbf{u}$. Naming the subspaces that answer questions of this shape is what Lecture 4 is for.

6 Periodic ends: the matrix $\mathbf{C}$

The last option is to let the configuration repeat: the same rod, the same loads, over and over, forever.



Periodic boundary conditions are a little bit contrived in this context of a stiff rod, but in general they are very useful for physics modelling. Because u and f repeat from cell to cell, the

grid closes up on itself: the neighbour to the left of the first node is the last node, and the neighbour to the right of the last node is the first.

Due to the periodicity, the finite differences **wrap around**:

$$\frac{1}{h^2} \begin{bmatrix} 2 & -1 & & -1 \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ -1 & & -1 & 2 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 2 & -1 & & -1 \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ -1 & & -1 & 2 \end{bmatrix}.$$

Every row still has the same three entries $-1, 2, -1$; the two rows that used to be short have had their missing -1 put back on the far side.

$\mathbf{C}$ is a **circulant matrix**: every row is the same, just shifted in position. We will discuss circulant matrices in much more detail during the Fourier section of the course, where that single structural fact turns out to diagonalise them completely.

A question worth asking now

Look at the row sums of $\mathbf{C}$. Do they tell you anything about whether $\mathbf{C}$ shares $\mathbf{B}$'s difficulty — and if so, what is the rod free to do this time?

7 The four matrices side by side

	End at $x = 0$	End at $x = 1$	First row	Last row
$\mathbf{K}$	fixed, $u(0) = 0$	fixed, $u(1) = 0$	2 -1	-1 2
$\mathbf{T}$	free, $u'(0) = 0$	fixed, $u(1) = 0$	1 -1	-1 2
$\mathbf{B}$	free, $u'(0) = 0$	free, $u'(1) = 0$	1 -1	-1 1
$\mathbf{C}$	periodic: the cell $[0, 1]$ repeats		2 -1 $\cdots$ -1	-1 $\cdots$ -1 2

Four physical situations, one interior stencil, four different pairs of boundary rows. $\mathbf{K}$ and $\mathbf{T}$ come from rods that are held somewhere; $\mathbf{B}$ comes from a rod that is held nowhere; $\mathbf{C}$ comes from a rod with no ends at all. Almost every matrix in this course is one of these four, a variant of one of them, or built out of them.

Where this is going

Next time we stop building matrices and start solving with them: Gaussian elimination, written as a factorisation $\mathbf{A} = \mathbf{LU}$. The two worked examples will be elimination on $\mathbf{T}$ and on $\mathbf{K}$ — and the symmetry we were careful to preserve today will show up there as $\mathbf{K} = \mathbf{LDL}^\top$.

Reading

Strang, *Computational Science and Engineering*, §1.1–1.2.